

Database/repo search

- After going through GITHUB for a while looking through various repos for a database of problems, I found something called the Natural Deduction pack by Alastair Carr. It contained exactly what I was looking for, a large question bank with hundreds of practical examples with solutions at the end.

[Natural-Deduction-Pack/Natural Deduction Pack.pdf at master · Alastair-Carr/Natural-Deduction-Pack · GitHub](#)

NaDeA

- While searching through various GitHub repositories, I stumbled upon NaDeA, which is a natural deduction assistant which uses Isabelle HOL as its logic backbone, it contained a .thy file which I thought would work perfectly in my project.
“The game’s inference engine is based on the formally verified natural deduction rules defined in NaDeA’s Isabelle theory file.”

[GitHub - logic-tools/nadea: NaDeA \(Natural Deduction Assistant\) - With a Formalization in Isabelle](#)

Weaknesses in Logiko’s mechanisms

Logiko’s drag-and-drop:

- unclear drop targets
- no ghost previews (what will happen if dropped)
- limited feedback on illegal moves
- minimal animation
- no rule-slot affordances
- poor scope visualization

Areas to improve upon

Slot-based rule application

- When you start dragging a formula, highlight valid rule application slots.

Ghost previews

- Show in advance where the inference line will appear.

Scope barriers

- When dragging into/out of subproofs, animate the scope border to show legality.

Ingredient-based rules

- Drag multiple premises into a “cauldron” to apply a rule (e.g., $\wedge E$ on a $\wedge$).

Summary

A review of existing natural-deduction tools revealed recurring weaknesses: limited visual engagement, poor feedback on illegal inference steps, and unclear subproof scope.

To address these issues, this project will use **React** to build a highly interactive, component-based interface capable of dynamically updating proof structures, highlighting rule affordances, and animating assumption scope changes.

To support intuitive interaction, the project will integrate **dnd-kit**, a modern drag-and-drop library that enables visual ghost previews, collision-based drop validation, and gesture-driven rule application.

Together, React and dnd-kit provide the necessary flexibility and responsiveness to overcome the UI shortcomings of existing systems and create a more engaging, accessible learning environment for natural-deduction proofs.

To ensure formal correctness and pedagogical coverage, this project draws on two established resources.

The **NaDeA** Isabelle theory file provides a verified specification of natural-deduction inference rules, informing the game’s internal proof-checking logic and preventing invalid derivations.

Complementing this, the **Natural Deduction Pack** offers a large, structured set of practice problems and worked examples across propositional and predicate logic. These materials serve as a foundation for level design, difficulty progression, and curriculum alignment, addressing the lack of curated exercise databases observed in many existing systems.